

	TECHNICAL SPECIFICATIONS HVAC WORKS	
Gujarat Biotechnology Research Centre	DOC.NO.GBRC_TS_HVAC_V.0	Doshi Consultants Pvt. Ltd.

18. INSULATION

The scope of this section comprises the supply and application of insulation to sheet metal ducting, piping & equipment conforming to the specifications.

All insulation on equipment, ducting and piping shall be applied only after the system has been pressure tested and for leak test satisfactorily.

DUCT INSULATION:

Insulation material shall be physically (irradiation) Cross-linked closed cell poly ethylene (XLPE) Thermal insulation backed with aluminium foil and Self Adhesive. Density of Material shall be 30Kg/m³, with having K value 0.032 W/m° K at 23°C Deg temp. Resistance of Fungi shall be Zero Growth. Operating Temperature Range at -40 to +115 Degreed Celsius. Physical property requirements at Complies (Type II-Sheet) of **ASTM C1427**. Flame & Smoke Complies OS ASTM E84/ UL 723.

Ducting insulation thickness shall be following as under:

Type of Duct	Insulation Thk
Supply Air Ducting	19 mm Thk.
Return Air Ducting	13 mm Thk.

External thermal insulation shall be provided as shown on drawing or identified in the bill of quantity.

Following procedure shall be adhered to for the Duct Insulation application:

Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work.

- Measurement of surface dimension shall be taken properly to cut closed cell elastomeric Rubber sheets to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade with straight cuts.
- Material shall be fitted under compression and no stretching of material shall be permitted. A thin coat of adhesive shall be applied on the back of the insulating material sheet and then on to the metal surface. When adhesive is tack dry, insulating material sheet shall be placed in position and pressed firmly to achieve a good bond. All longitudinal and transverse joints shall be sealed by providing 6 mm thick 50 mm wide self-adhesive insulation tape. The adhesive shall be strictly as recommended by the manufacturer.



Complies with all specified requirements except ASTM E84. Instead, it meets Class 0 fire performance as per BS 476 Part 6 and Part 7, which are recognized standards for fire propagation and surface spread of flame.

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CHILLED WATER PIPING INSULATION:

Manifold piping and valve will be insulated with PUF insulation with aluminum cladding and low thermal conductivity.

Thickness of the insulation will be as specified for the individual application. Adhesive used for setting the insulation will be non-flammable, vapour proof adhesive.

General Recommended thickness of insulation

CHILLED WATER (CHW)				
Pipe size	Insulation Thk		Pipe size	Insulation Thk
NB	mm		NB	mm
15	25		100	32
20	25		125	32
25	25		150	32
32	32		200	32
40	32		250	38
50	32		300	38
65	32		350	38
80	32		400	38

All chilled water and drain piping shall be insulated in the manner as specified herein.

Insulation shall be applied only after the piping system has been satisfactorily tested for leaks at 2.5 times the working pressure or at minimum 10 Kg/Sqm. test pressure. All surfaces to be insulated shall be brushed, cleaned of all foreign materials such as scale, dust, dirt, mortar and oil removed. One coat of zinc chromate primer shall be applied and then allowed to dry before application of insulation.

After cleaning and application of one coat of primer paint, two coats of insulation adhesive shall be applied. The insulation adhesive shall be either hot bitumen 85/25 or 85/40 (IS 702) or CPRX compound of approved make applied uniformly in the ratio of not less than 1.5 Kg/Sqm.

Noted

A similar layer of adhesive shall be applied on the inside surface of insulation. Insulation material shall be stuck to the surface with joints staggered. The adjoining sections shall be tightly pressed together.

All joints shall be sealed with adhesive material. Voids if any, shall be packed with suitable cut pieces of insulation material.

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Insulation material shall be wrapped with 400 g polythene sheets with minimum 4 inches overlapping for vapour barrier. Alternatively, tissue paper with two coats of bitumen on both sides may also be used. PVC straps at 400 mm centre shall be used to hold insulation and vapour barrier together.

Then sealt of equivalent make cladding, the surface covering shall be made from a 3-layer composite film of metallic appearance with aluminum intermediate layer and UV protection. Material shall be 350µm in thickness, with a total weight ≥ 500 g/m², and on pipe sections have a permanent curl. The material shall be Class B (GB 8624), Class 0 (BS 476) and have a Flammability Index ≤ 5 (AS1530.2).

Cladding shall be installed with 50mm overlaps on all horizontal and longitudinal joins. On outdoor applications, joints along the material length shall be installed facing downward. The joints shall be secured with plastic rivets in the overlap area, installed every 200mm along the material length.

Manufacturer recommended adhesive can be used for flat sheets, or where restricted access prevents use of rivets. All joints and overlaps must be finished with manufacturer recommended PSA Tape.

All bends, T-sections and encaps shall preferably be clad with thermoformed fittings of composite covering system. For fittings of an unusual shape, or large size, the cladding material should be cut and installed as per traditional metal cladding.